THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1001	Divide 1000 by the number and round up.
2	**(d)** 19	Total numbers − primes − 1 (1 is neither).
3	**(b)** 24	LCM.
4	**(c)** 24	HCF of three numbers.
5	**(d)** 5	Digit sum must be a multiple of 9.
6	**(a)** 997	Subtract the remainder first.
7	**(d)** 41	List the primes after 10.
8	**(b)** 97	Check numbers just below.
9	**(a)** Both A and R are true and R is the correct explanation of A.	A prime has exactly two factors and a composite has more than two; 1 has just one factor.
10	**(a)** Both A and R are true and R is the correct explanation of A.	So every even number above 2 has at least three factors and is composite.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	LCM of 8, 12, 18 (1); $= 72$ (1).	1+1
12	The numbers include a multiple of 3 (1); Hence the product is divisible by 3 (1).	1+1
13	$84 = 2^2 \times 3 \times 7$ (1); Needed multiplier: 15 (1).	1+1
14	LCM of 10, 14, 35 (1); $= 70$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM of 48, 72, 108 is 432 seconds (1); That is 7 minutes 12 seconds (1); Time: 8:07:12 am (1).	1+1+1
16	Subtract the remainder: 72 and 96 (1); HCF of 72 and 96 is 24 (1); The number is 24 (it must exceed 4) (1).	1+1+1
17	LCM $= 24$ (1); Required number $= LCM + 5 = 29$ (1); Check each remainder (1).	1+1+1
	OR: LCM of 30, 45, 60 is 180 seconds (1); That is 3 minutes 0 seconds (1); Time: 8:03:00 am (1).	1+1+1
18	Subtract the remainder: 228 and 276 (1); HCF of 228 and 276 is 12 (1); The number is 12 (it must exceed 4) (1).	1+1+1
	OR: LCM $= 60$ (1); Required number $= LCM + 3 = 63$ (1); Check each remainder (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $12 = 2^2 \times 3$, $18 = 2 \times 3^2$ (1). (ii) LCM $= 2^2 \times 3^2 = 36$ (1). (iii) 9:36 am (1). (iv) 11:00 am is 120 minutes after 9:00; $120 \div 36$ gives 3 more times, at 9:36, 10:12 and 10:48 am (2).	1+1+1+2
20	(i) Side $=$ HCF $= 24$ m (1); (ii) Beds $= 3 \times 2 = 6$ (1); (iii) Perimeter of one bed $= 96$ m (1); Total fencing $= 576$ m (1); Cost $= ₹2304$ (1).	1+1+1+1+1
	OR: LCM $= 90$ (1); Smallest 4-digit multiple: 1080 (1); Largest 4-digit multiple: 9990 (1); Verify divisibility (2).	1+1+1+2

Mistakes to avoid:

- Calling 1 a prime number; 1 is neither prime nor composite.
- Thinking all odd numbers are prime (9, 15, 21, 91 are composite) or that 2 is not prime.
- Taking the greatest powers for HCF or the smallest powers for LCM; it is the other way round.
- Using LCM when the question asks for the largest equal size (HCF), and the reverse for 'meet again'.
- Applying the divisibility test for 4 or 8 to the digit sum instead of the last two or three digits.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 34	Total numbers − primes − 1 (1 is neither).
2	**(b)** 60	LCM.
3	**(a)** 30	HCF of three numbers.
4	**(a)** 2	Digit sum must be a multiple of 9.
5	**(c)** 994	Subtract the remainder first.
6	**(d)** 60	List the primes after 10.
7	**(b)** 149	Check numbers just below.
8	**(b)** 35	$385 = 5 \times 7 \times 11$
9	**(c)** A is true but R is false.	91 = 7 x 13 so A is true. 91 is not divisible by 2, 3 or 5 but is composite, so R is false.
10	**(d)** A is false but R is true.	51 = 3 x 17, so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The numbers include a multiple of 4 (1); Hence the product is divisible by 4 (1).	1+1
12	$66 = 2 \times 3 \times 11$ (1); Needed multiplier: 20 (1).	1+1
13	LCM of 9, 15, 21 (1); $= 315$ (1).	1+1
14	The numbers include a multiple of 3 (1); Hence the product is divisible by 3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM of 36, 60, 90 is 180 seconds (1); That is 3 minutes 0 seconds (1); Time: 8:03:00 am (1).	1+1+1
16	Subtract the remainder: 88 and 112 (1); HCF of 88 and 112 is 8 (1); The number is 8 (it must exceed 7) (1).	1+1+1
17	LCM $= 72$ (1); Required number $= LCM + 4 = 76$ (1); Check each remainder (1).	1+1+1
	OR: LCM of 40, 50, 100 is 200 seconds (1); That is 3 minutes 20 seconds (1); Time: 8:03:20 am (1).	1+1+1
18	Subtract the remainder: 198 and 240 (1); HCF of 198 and 240 is 6 (1); The number is 6 (it must exceed 2) (1).	1+1+1
	OR: LCM $= 60$ (1); Required number $= LCM + 7 = 67$ (1); Check each remainder (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $120 = 2^3 \times 3 \times 5$, $84 = 2^2 \times 3 \times 7$ (1). (ii) HCF $= 2^2 \times 3 = 12$ cm (1). (iii) $120 \div 12 = 10$ along the length and $84 \div 12 = 7$ along the breadth (1). (iv) $10 \times 7 = 70$ tiles (1); check: $\frac{120 \times 84}{12 \times 12} = \frac{10080}{144} = 70$ (1).	1+1+1+2
20	(i) Side $=$ HCF $= 18$ m (1); (ii) Beds $= 5 \times 3 = 15$ (1); (iii) Perimeter of one bed $= 72$ m (1); Total fencing $= 1080$ m (1); Cost $= ₹3240$ (1).	1+1+1+1+1
	OR: LCM $= 336$ (1); Smallest 4-digit multiple: 1008 (1); Largest 4-digit multiple: 9744 (1); Verify divisibility (2).	1+1+1+2

Mistakes to avoid:

- Calling 1 a prime number; 1 is neither prime nor composite.
- Thinking all odd numbers are prime (9, 15, 21, 91 are composite) or that 2 is not prime.
- Taking the greatest powers for HCF or the smallest powers for LCM; it is the other way round.
- Using LCM when the question asks for the largest equal size (HCF), and the reverse for 'meet again'.
- Applying the divisibility test for 4 or 8 to the digit sum instead of the last two or three digits.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 36	LCM.
2	**(a)** 42	HCF of three numbers.
3	**(c)** 8	Digit sum must be a multiple of 9.
4	**(b)** 995	Subtract the remainder first.
5	**(c)** 83	List the primes after 10.
6	**(c)** 199	Check numbers just below.
7	**(c)** 10	$910 = 2 \times 5 \times 7 \times 13$
8	**(a)** 1012	Divide 1000 by the number and round up.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	HCF(15, 28) = 1, so A is true. R is true but does not explain why they are co-prime.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Any multiple of 9 is a multiple of a multiple of 3, so it is divisible by 3.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$90 = 2 \times 3^2 \times 5$ (1); Needed multiplier: 77 (1).	1+1
12	LCM of 12, 16, 24 (1); $= 48$ (1).	1+1
13	The numbers include a multiple of 5 (1); Hence the product is divisible by 5 (1).	1+1
14	$105 = 3 \times 5 \times 7$ (1); Needed multiplier: 44 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM of 24, 40, 60 is 120 seconds (1); That is 2 minutes 0 seconds (1); Time: 8:02:00 am (1).	1+1+1
16	Subtract the remainder: 209 and 242 (1); HCF of 209 and 242 is 11 (1); The number is 11 (it must exceed 6) (1).	1+1+1
17	LCM $= 90$ (1); Required number $= LCM + 2 = 92$ (1); Check each remainder (1).	1+1+1
	OR: LCM of 45, 75, 90 is 450 seconds (1); That is 7 minutes 30 seconds (1); Time: 8:07:30 am (1).	1+1+1
18	Subtract the remainder: 105 and 147 (1); HCF of 105 and 147 is 21 (1); The number is 21 (it must exceed 8) (1).	1+1+1
	OR: LCM $= 84$ (1); Required number $= LCM + 6 = 90$ (1); Check each remainder (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 10 primes (2, 3, 5, 7, 11, 13, 17, 19, 23, 29) (1). (ii) 23 and 29 (1). (iii) No, $27 = 3 \times 9$ has more than two factors (1). (iv) The digit sum of 57 is $5 + 7 = 12$, which is divisible by 3, so 57 is divisible by 3 (1); $57 = 3 \times 19$, so it is composite (1).	1+1+1+2
20	(i) Side $=$ HCF $= 12$ m (1); (ii) Beds $= 9 \times 7 = 63$ (1); (iii) Perimeter of one bed $= 48$ m (1); Total fencing $= 3024$ m (1); Cost $=$ ₹15120 (1).	1+1+1+1+1
	OR: LCM $= 360$ (1); Smallest 4-digit multiple: 1080 (1); Largest 4-digit multiple: 9720 (1); Verify divisibility (2).	1+1+1+2

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- Taking the greatest powers for HCF or the smallest powers for LCM; it is the other way round.
- Using LCM when the question asks for the largest equal size (HCF), and the reverse for 'meet again'.
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